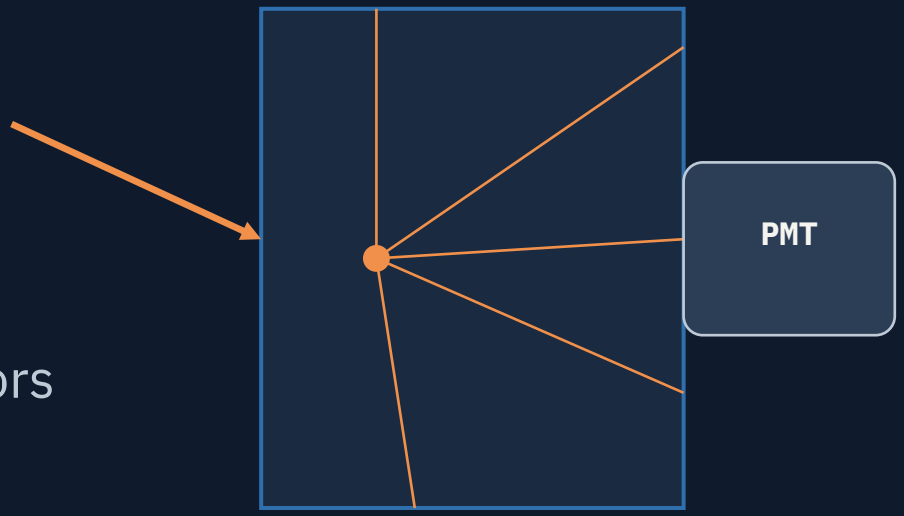
PARTICLE DETECTOR DESIGN AND SIMULATION

Scintillation Detectors



Light-Producing Materials and Light-Counting Photodetectors

Introduction · how it works · organic and inorganic types · light collection · Geant4 · PMT · SiPM

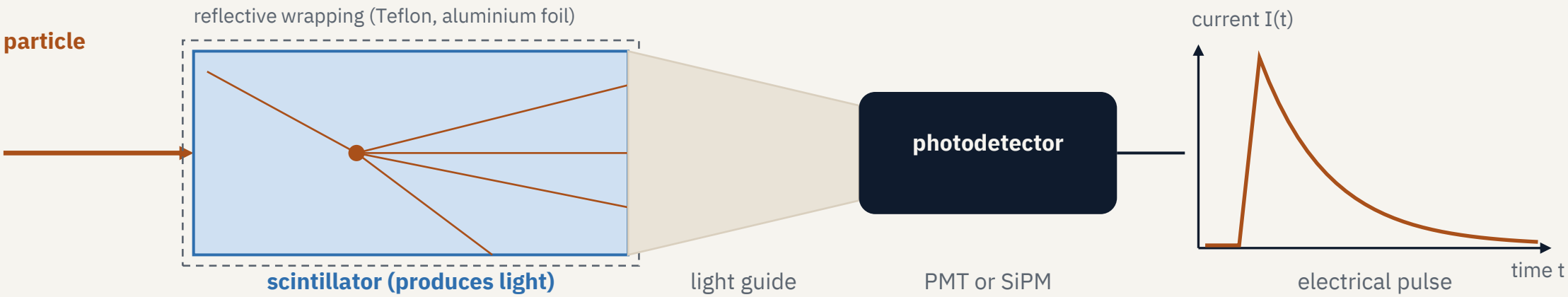
Kaan Yüksel OYULMAZ · Ankara University Institute of Accelerator Technologies · 2026

SCINTILLATION DETECTORS

What will we cover?

01	**Introduction**	What scintillation is, the parts of the detector, how the signal forms
02	**How it works**	Light yield, number of photons, decay time, energy resolution
03	**Organic scintillators**	Molecular levels, fluorescence, plastic and liquid scintillators
04	**Inorganic scintillators**	Band structure, activator, NaI(Tl), CsI(Tl), BGO, LYSO, $PbWO_4$
05	**Light collection**	Total internal reflection, reflector, light guide, wavelength shifter
06	**Design in Geant4**	BC-408 plastic slab, optical properties, photon counting
07	**PMT**	Photocathode, dynodes, gain, high voltage
08	**SiPM**	Pixel array in Geiger mode, gain, noise, saturation
09	**Comparison**	PMT and SiPM side by side
10	**PMT in Geant4**	Same code, steps E, F, G: window and sensitive detector, hit, output signal

What is scintillation? Parts of the detector



- **Scintillation:** A small part of the deposited energy (a few percent) is given back as visible or ultraviolet light.
- **Four parts:** Scintillator, reflective wrapping, light guide, photodetector.
- **Signal:** A current pulse that rises in nanoseconds and decays over a material-specific time. The area under it is proportional to the deposited energy.

Numerical example: 1 MeV deposited energy

NaI(Tl) crystal: 38/keV × 1000 keV = **38 000 photons**

Plastic scintillator: 10/keV × 1000 keV = **10 000 photons**

Photon energy ≈ 3 eV: 38 000 × 3 eV ≈ 114 keV, only 11% of the energy becomes light, the rest is heat.

How many photons, and how fast does the light decay?

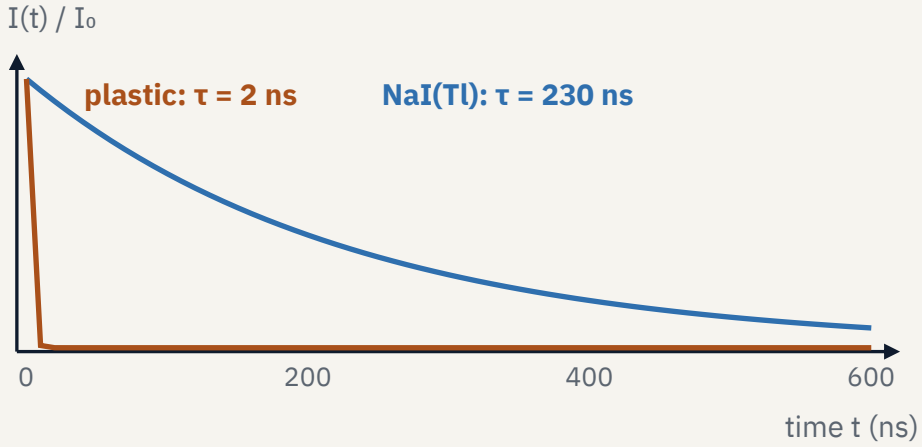
Number of photons

$$N_{photon} = L_Y \times E_{deposited}$$

N_{photon}: number of photons produced

L_Y: light yield, photons / keV (NaI(Tl) 38, plastic 10, BGO 8)

$E_{deposited}$: energy deposited by the particle in the scintillator, keV



Decay of the light with time

$$I(t) = I_0 \cdot e^{-t/\tau}$$

I(t): light intensity emitted at time t

I_0: initial intensity (t = 0)

τ (tau): decay time, the mean emission time: in this time the light drops to about one third (37%) of its initial value

Why does the light decay with time?

Emission happens in individual molecules. Some excited molecules emit right away, others later. When all their emissions add up, the light is strong at first and then fades. The mean emission time is τ.

What do we expect from a good scintillator?

- **High light yield:** More photons, better resolution.
- **Short decay time τ:** Fast counting, good timing, less pile-up.
- **What sets τ?** The emitter: the molecule in organics (plastic 2 ns), the activator in inorganics (Ce^{3+} 40 ns, Tl^{+} 230 ns).
- **Transparent to its own light:** Must not reabsorb its own photons.
- **Suitable wavelength:** Where the photodetector is most sensitive, ≈ 400 nm.
- **Density and Z:** High, to stop gamma rays.
- **Practical:** Not hygroscopic, machinable, cheap.

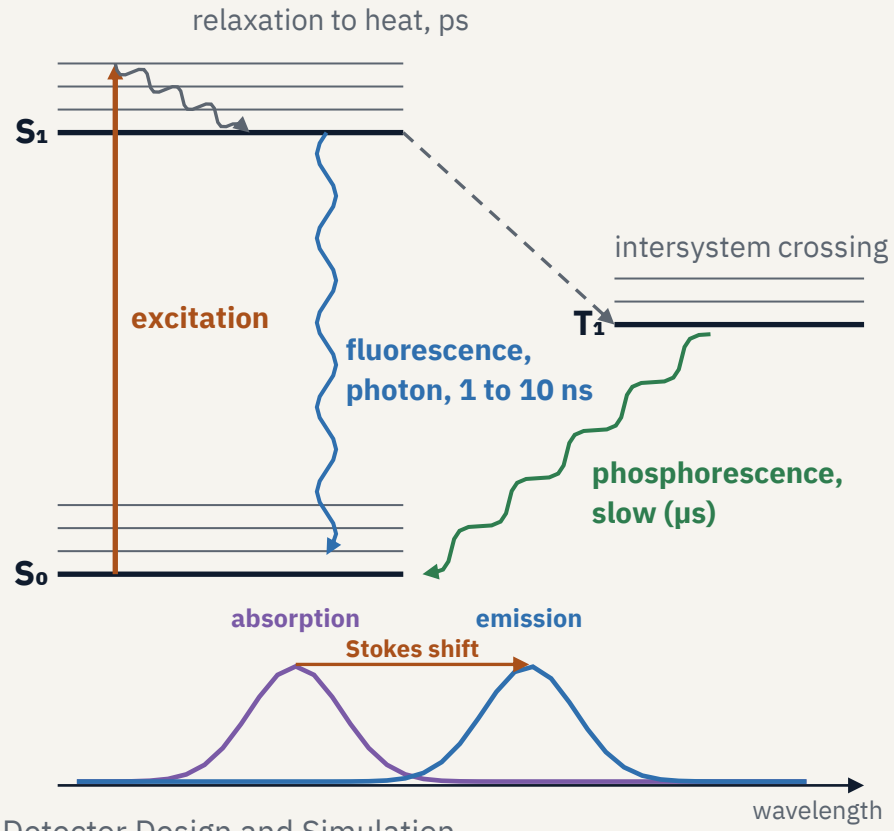
Material	Yield (ph/keV)	τ (ns)	λ (nm)	ρ (g/cm³)	Moisture
NaI(Tl)	38	230	415	3.67	yes
CsI(Tl)	54	1000	550	4.51	slightly
BGO	8	300	480	7.13	no
LYSO(Ce)	30	40	420	7.1	no
$PbWO_4$	0.2	10	420	8.28	no
Plastic	10	2	425	1.03	no
Liquid (NE-213)	12	3	425	0.87	no

τ, λ, ρ: decay time, wavelength of the emitted light, density

Moisture: absorbs water from the air? NaI(Tl) is kept in a sealed can

> **Choice:** NaI(Tl) or LYSO to measure energy, plastic for fast timing, BGO or $PbWO_4$ to stop gammas.

Organic scintillators: The molecule itself emits



- **Source:** The π electrons of benzene-ring molecules. Crystal, liquid or plastic, the mechanism is the same.
- **Fluorescence:** A photon on the return from S_1 to S_0, 1 to 10 ns. This is where fast counting comes from.
- **Phosphorescence:** Slow emission via the triplet T_1, used in liquids to tell neutrons from gammas.

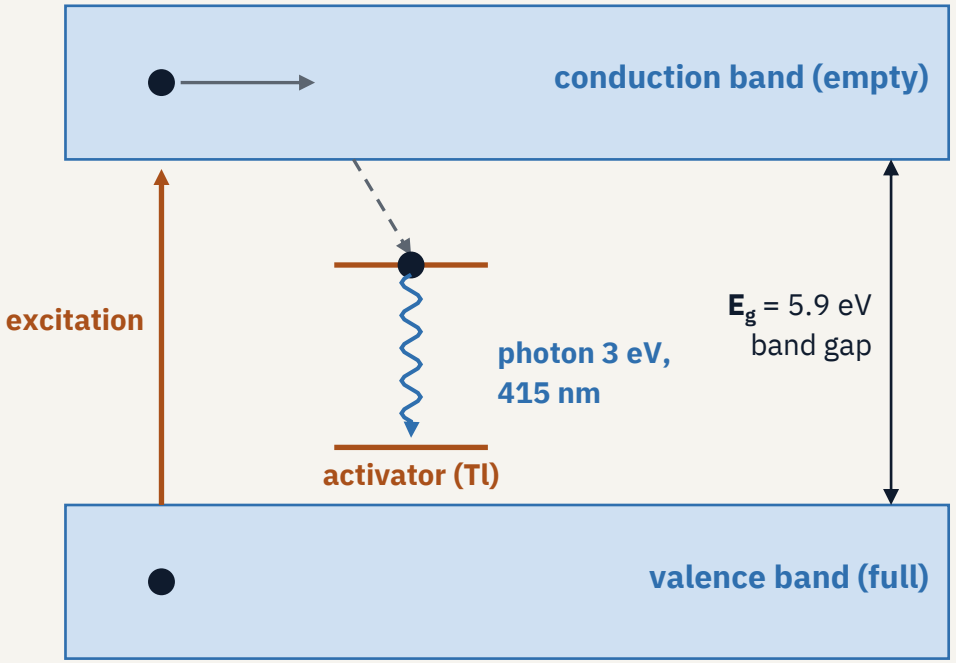
Stokes shift: Why is the emitted light not reabsorbed?

Absorption happens at high energy (PPO: 303 nm), **emission** at lower energy after relaxation to heat (PPO: 365 nm).

Result: Emission lies outside the absorption band, so the material is transparent to its own light.

Type	Example	τ (ns)	photons/keV
Crystal	anthracene, stilbene	30	17
Liquid	NE-213 (xylene + PPO)	3	12
Plastic	polystyrene + PPO + POPOP	2	10

Inorganic crystals: Band structure and activator



Photon energy (3 eV) is below the band gap (5.9 eV): the crystal is transparent to its own light

- **Source:** The particle lifts an electron from the valence to the conduction band.
- **Problem:** A direct return gives a 5.9 eV photon that the crystal reabsorbs at once.
- **Solution, the activator:** A few foreign atoms (0.1% Tl in NaI) open levels in the gap: emission at 3 eV, the crystal is transparent.
- **Why for gammas:** Dense and high Z.

Why do BGO and $PbWO_4$ emit without an activator? The emitting ion is part of the lattice itself (Bi^{3+}, WO_4^{2-}) and its level lies inside the band gap. The price is little light.

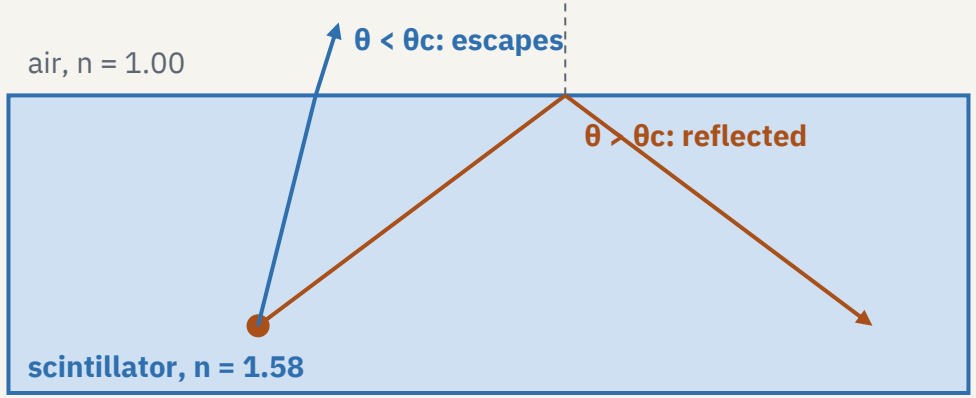
The activator sets τ: Tl^+ is slow, NaI(Tl) 230 ns. Ce^{3+} is fast, LYSO(Ce) 40 ns. The symbol in parentheses is the activator.

Organic or inorganic?

Property	Organic (plastic, liquid, anthracene)	Inorganic (NaI(Tl), CsI(Tl), BGO, LYSO)
Source of the light	π electrons of a single molecule	Crystal lattice and activator levels
Decay time	1 to 10 ns: fast	40 to 1000 ns: slow
Light yield	10 to 17 photons/keV	8 to 54 photons/keV
Density, Z	≈ 1 g/cm³, low Z (C, H)	3.7 to 8.3 g/cm³, high Z (I, Cs, Bi, Pb)
For gammas	Poor (Compton dominates, no photopeak)	Good (photoelectric peak visible)
Shape, price	Cut to any size, cheap	Grown as a crystal, expensive
Robustness	Not hygroscopic, yellows under radiation	NaI is hygroscopic, needs a sealed can
Typical use	Trigger, time of flight, hodoscope, neutrons	Gamma spectroscopy, PET, calorimeter

> **Rule:** If the question is "how many particles, when?" use plastic; if it is "a gamma of what energy?" use NaI(Tl) or LYSO.

How does the light reach the photodetector?



Critical angle

$$\sin \theta_c = n_{air} / n_{scintillator}$$

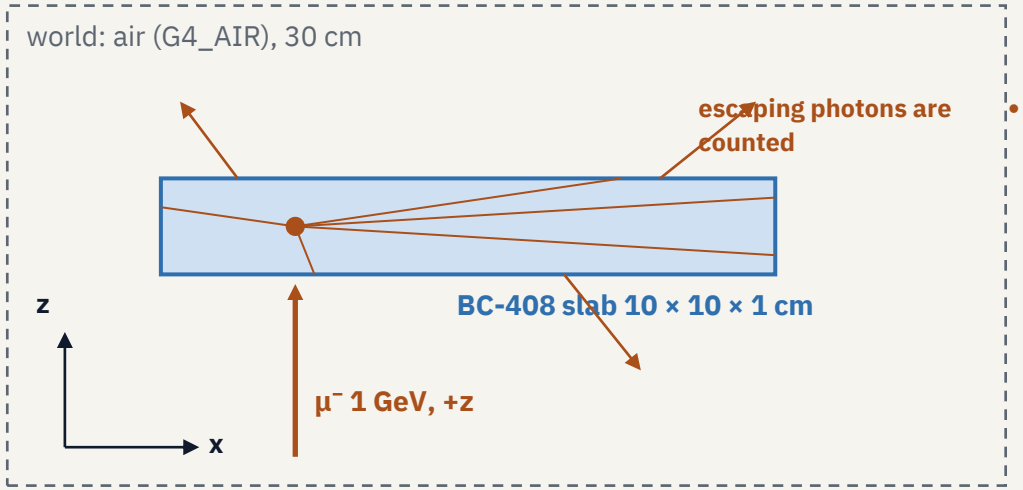
θ_c: critical angle; a ray hitting the surface at a larger angle to the normal cannot get out

n: refractive index, air 1.00, plastic 1.58, NaI 1.85

Example, plastic: sin θc = 1 / 1.58 = 0.633, θc = 39°. Every ray more grazing than 39° stays inside.

- **Total internal reflection:** Light more grazing than the critical angle travels by internal reflection. The scintillator is its own light guide.
- **Reflective wrapping:** Teflon tape or aluminium foil sends escaping light back. Light-tight black tape on top.
- **Light guide:** A transparent Plexiglas piece that couples a large scintillator face to a small photodetector.
- **Wavelength-shifting fibre:** Absorbs the blue light and carries it along the fibre as green. Brings the light of large slabs to a single point.
- **Optical grease:** No air gap between the scintillator and the photodetector glass. Index matching reduces reflection losses.
- **Result:** Typically 30% to 60% of the produced photons reach the photodetector.

Scintillator design in Geant4: BC-408 plastic slab



A	**DetectorConstruction** BC-408 optical properties, n for air	No photons, histogram empty
B	**PhysicsList** G4OpticalPhysics: scintillation, absorption, reflection	Photons are produced, not counted
C	**SteppingAction** Count created and escaping photons	≈ 18 000 produced, 66% escape
D	**DetectorConstruction** Reflective wrapping: 95% reflecting surface	Escaping: 0

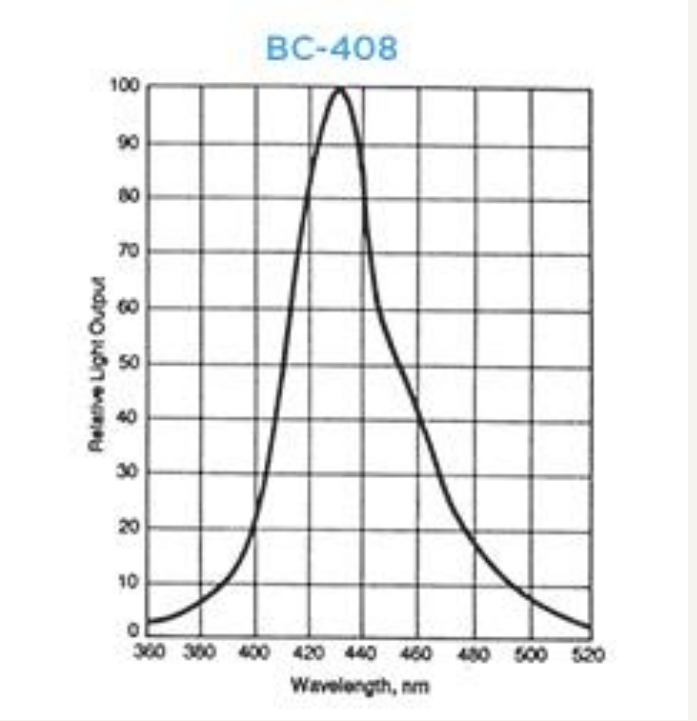
- **Already there:** Geometry (air-filled world, BC-408 slab), source, energy histogram. The code compiles and runs.
- **Commented out:** Seven steps, each line starting with //. A to D here, E to G in the PMT section (10). In class we uncomment them one by one, compile, run and look at the histogram.

	BC-400	BC-404	BC-408	BC-412	BC-416
Principal Uses/Applications	general purpose	fast counting	TOF large area	large area	large area economy
Scintillation Properties					
Light Output, %Anthracene	65	68	64	60	38
Rise Time, ns	0.9	0.7	0.9	1.0	-
Decay Time (ns)	2.4	1.8	2.1	3.3	4.0
Pulse Width, FWHM, ns	2.7	2.2	~2.5	4.2	5.3
Wavelength of Max. Emission, nm	423	408	425	434	434
Light Attenuation Length, cm*	160	140	210	210	210
Bulk Light Attenuation Length, cm	250	160	380	400	400
Atomic Composition					
No. H Atoms per cc ($\times 10^{22}$)	5.23	5.21	5.23	5.23	5.25
No. C Atoms per cc ($\times 10^{22}$)	4.74	4.74	4.74	4.74	4.73
Ratio H:C Atoms	1.103	1.100	1.104	1.104	1.110
No. of Electrons per cc ($\times 10^{23}$)	3.37	3.37	3.37	3.37	3.37

Saint-Gobain datasheet: BC-408 column, values used in the code

Step A: Optical properties of BC-408

```
// --- STEP A: optical properties (leading // removed) ---
std::vector<G4double> energy = {2.48 * eV, 2.76 * eV,
                                3.92 * eV, 3.10 * eV};  // 500, 450,
425, 400 nm
G4MaterialPropertiesTable* mpt = new G4MaterialPropertiesTable();
mpt->AddProperty("RINDEX", energy, {1.58, 1.58, 1.58, 1.58});
mpt->AddProperty("ABSLENGTH", energy,
                 {380.*cm, 380.*cm, 380.*cm, 380.*cm});   // bulk
mpt->AddProperty("SCINTILLATIONCOMPONENT1", energy,
                 {0.10, 0.60, 1.00, 0.25});   // from the curve
mpt->AddConstProperty("SCINTILLATIONYIELD", 10000. / MeV);
mpt->AddConstProperty("RESOLUTIONSCALE", 1.0);
mpt->AddConstProperty("SCINTILLATIONTIMECONSTANT1", 2.1 * ns);
plastic->SetMaterialPropertiesTable(mpt);
// --- end of STEP A ---
```



From the datasheet to the code: Light yield 64% of anthracene = 10 000 photons/MeV. Decay 2.1 ns. Bulk attenuation 380 cm. n = 1.58.

Emission spectrum: Four points read off the curve: 400 nm 25%, 425 nm 100%, 450 nm 60%, 500 nm 10%. Geant4 uses them as weights.

> **Check:** Compile, ./ders sint.mac. The energy histogram fills (≈ 1.9 MeV), the photon histogram is empty: no optical processes yet.

Steps B and C: Optical physics and photon counting

step B: optical physics

```
// PhysicsList.cc, constructor
RegisterPhysics(new G4OpticalPhysics());
```

step C: photon counting

```
// SteppingAction.cc
if (track->GetDefinition() ==
    G4OpticalPhoton::OpticalPhotonDefinition())
{
  if (track->GetCurrentStepNumber() == 1)
    fEvent->AddCreated();     // created
  if (step->GetPostStepPoint()->GetStepStatus()
      == fWorldBoundary)
    fEvent->AddEscaped();     // escaped
  return;   // not added to the energy sum
}
// EventAction.hh (given):
void AddCreated() { fNcreated += 1; }
void AddEscaped() { fNescaped += 1; }
```

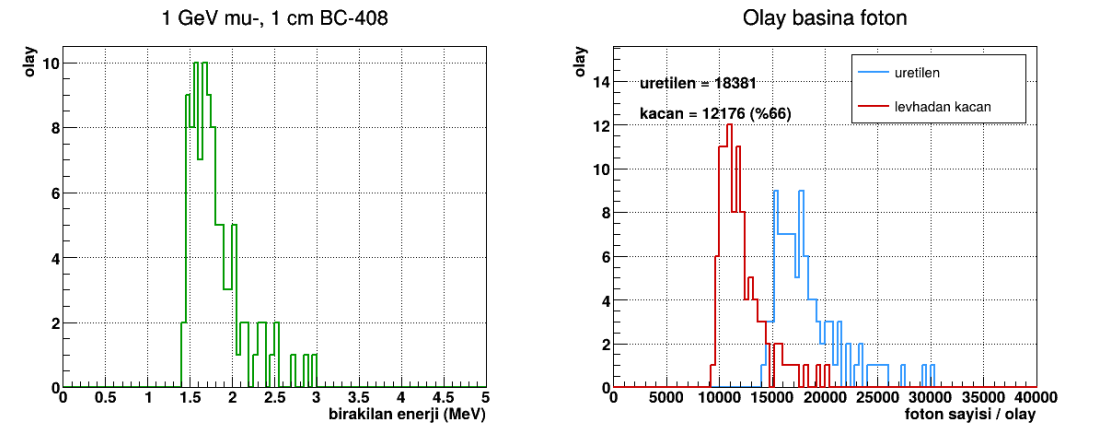
- **G4OpticalPhysics:** Scintillation, Cherenkov, absorption, reflection and refraction at surfaces. All with one line.
- **What is an optical photon?** A separate particle type in Geant4: opticalphoton. Each one is tracked individually; there are many.
- **Created photon:** The first step of a track (GetCurrentStepNumber() == 1) comes exactly once per photon.
- **Escaping photon:** A photon whose step ends at the world boundary (fWorldBoundary) has left the slab. Reflected photons never reach it.
- **Counters:** fNcreated and fNescaped in EventAction, filled into two histograms at event end.

Check A+B: Photons are produced but not counted, histograms empty. The run is slower: 18 000 photons tracked per event.

Check A+B+C: Per event **18 000 photons are produced, 12 000 escape (66%)**. The rest are trapped inside and absorbed.

Step D and the result: Reflective wrapping

```
step D: reflective wrapping
// DetectorConstruction.cc, after the slab is placed
G4OpticalSurface* wrap = new G4OpticalSurface("Wrap");
wrap->SetType(dielectric_metal);   // reflect or absorb
wrap->SetModel(unified);
wrap->SetFinish(polished);
G4MaterialPropertiesTable* mptWrap = new
G4MaterialPropertiesTable();
mptWrap->AddProperty("REFLECTIVITY", energy,
{0.95,0.95,0.95,0.95});
wrap->SetMaterialPropertiesTable(mptWrap);
new G4LogicalBorderSurface("WrapSurface", physSlab,
physWorld, wrap);
```



Case	Produced	Escaped	Fraction
A + B + C (no wrapping)	18 000	12 000	66%
A + B + C + D (wrapped)	19 000	0	0%

- **dielectric_metal:** The surface behaves like a metal: the photon is reflected with 95% probability, absorbed with 5%.
- **G4LogicalBorderSurface:** The surface is attached to the boundary between the slab and the world.

Try: Set REFLECTIVITY to 0.98. Make the slab 20 cm long. Leave one face unwrapped. How does the number of escaping photons change in each case? Unwrapped plot: blue produced, red escaped.

From light to electricity: Photoelectric effect and QE

Photoelectric effect

$$E_{electron} = h\nu - W$$

hv: photon energy, h Planck's constant, v frequency. 2.95 eV for 420 nm

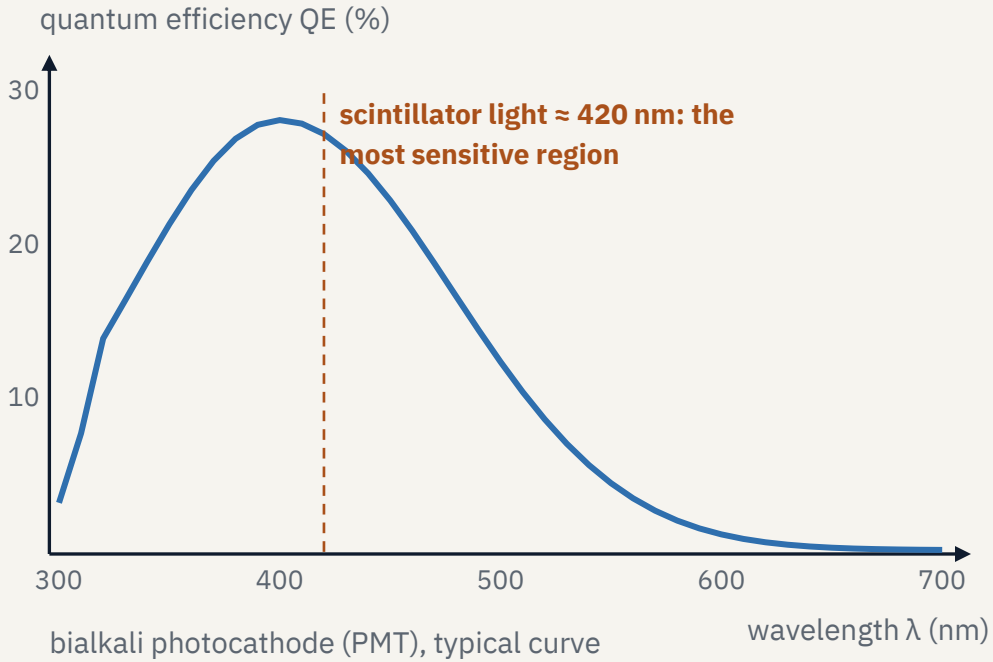
W: work function: the minimum energy needed to free an electron from the surface (bialkali ≈ 2 eV)

$E_{electron}$: kinetic energy of the freed electron; if negative, no electron comes out

Quantum efficiency (QE)

$$QE = N_{photoelectron} / N_{photon}$$

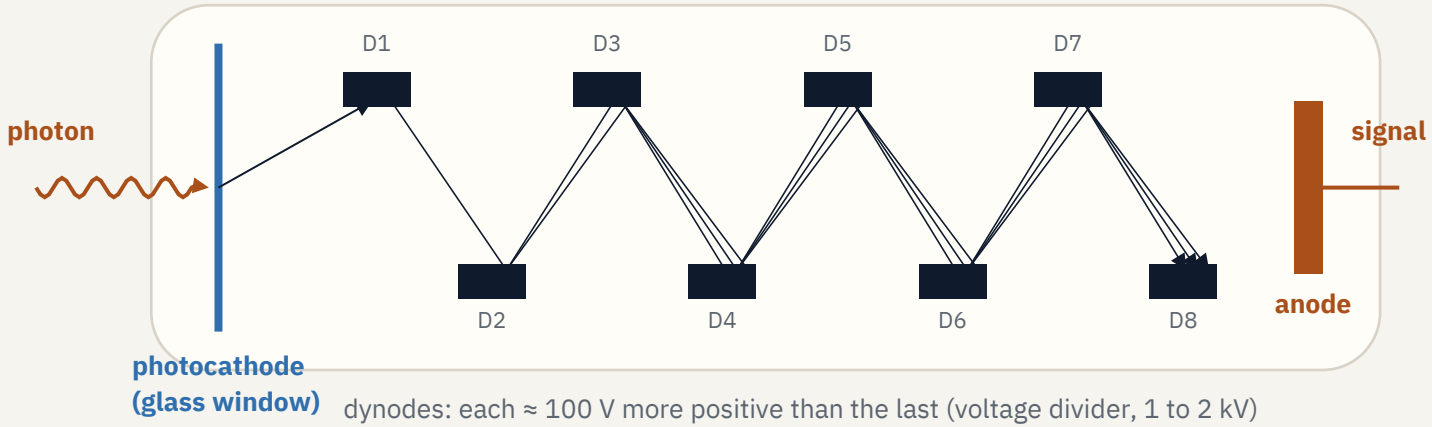
How many of 100 photons free an electron. Bialkali PMT: at 400 nm **25% to 30%**, under 5% at 600 nm.



bialkali photocathode (PMT), typical curve

Why multiplication is needed: A single photoelectron carries 1.6×10^{-19} C, which cannot be measured directly. The PMT with dynodes and the SiPM with an avalanche multiply this charge a million times.

Photomultiplier tube (PMT): Multiplication with dynodes



dynodes: each ≈ 100 V more positive than the last (voltage divider, 1 to 2 kV)

- **Photocathode:** The photon frees an electron here (QE ≈ 25%).
- **Dynodes:** At each impact the number of electrons grows δ times; with 10 dynodes, a million times.
- **Anode:** The multiplied electrons are collected and come out as a current pulse.
- **High voltage:** 1 to 2 kV; the gain changes very steeply with voltage, a stable supply is essential.

Gain

$$G = \delta^n$$

G: gain: electrons at the anode / electrons from the photocathode

δ (delta): electrons released per impact, 2 to 5, grows with voltage

n: number of dynodes, 8 to 12. Example: δ = 4, n = 10: $G = 4^{10} \approx 10^6$

Numerical example: A single photoelectron

Charge: $Q = G \cdot e = 10^6 \times 1.6 \times 10^{-19}$ C = **0.16 pC**

Current: if the pulse lasts 5 ns, I = Q / t = 32 µA

Voltage: across 50 Ω, V = I · R ≈ **1.6 mV**: a single photon is measurable

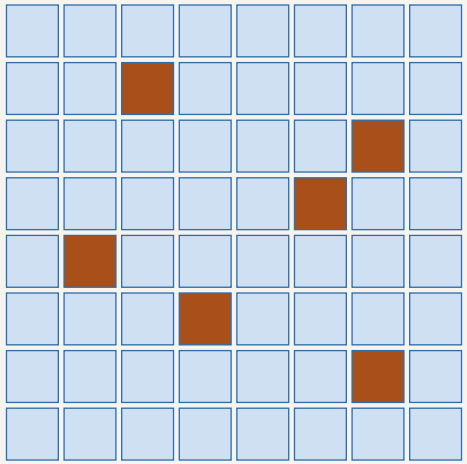
PMT: Properties, strengths and limits

Property	Typical value	Meaning
Quantum efficiency QE	25% to 30% (400 nm)	A quarter of the incoming photons become electrons
Gain G	10^5 to 10^7	A single photon becomes visible
High voltage	1 to 2 kV	Gain is very sensitive to voltage, a stable supply is needed
Rise time	1 to 3 ns	Fast timing, sub-nanosecond time resolution
Dark current	100 to 1000 counts/s	Thermal electrons with no light, small
Photocathode diameter	1 to 50 cm	A large area is read out with a single tube
Magnetic field	Even a few gauss disturb it	Does not work inside a magnet, needs mu-metal shielding

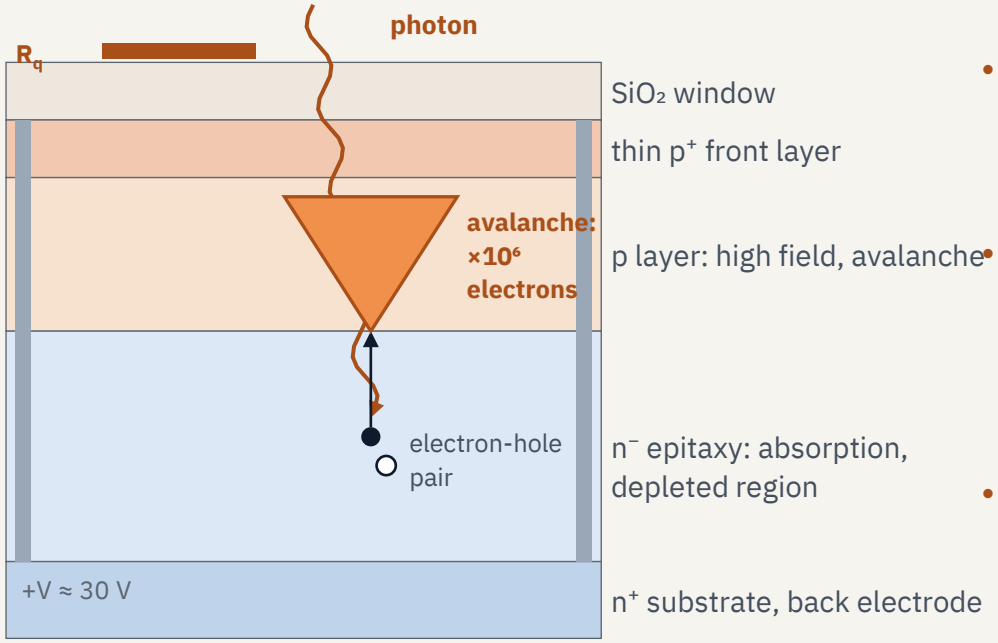
Strengths: Very high gain, very low noise, large area, fast, proven technology.

Limits: Fragile glass vacuum tube, bulky, high voltage, sensitive to magnetic fields, QE limited to about 25%.

Silicon photomultiplier (SiPM): Geiger-mode pixel array



pixel array: 3600 pixels of 50 µm on a 3 × 3 mm² chip. Orange: 6 fired pixels = 6 photons



grey: isolation trench, separates it from the neighbouring pixel

- **Pixel:** An avalanche photodiode of 10 to 75 µm, hundreds or thousands per mm².
- **Geiger mode:** 2 to 5 V above breakdown. One photon starts a full avalanche: the pixel fires or it does not.
- **Quenching:** The series resistor cuts the current, the avalanche stops, the pixel recovers in ≈ 100 ns.
- **Output:** Pixels in parallel: the signal is proportional to the number of fired pixels.

Gain per pixel ≈ 10^6: with 30 V as much as a PMT, immune to magnetic fields, far smaller

SiPM: Overvoltage, noise and saturation

Overvoltage

$$\Delta V = V_{operating} - V_{breakdown}$$

Example: 30 V – 27 V = 3 V overvoltage

$V_{breakdown}$: the voltage at which the avalanche starts, 25 to 50 V; drifts by 20 to 50 mV per degree with temperature

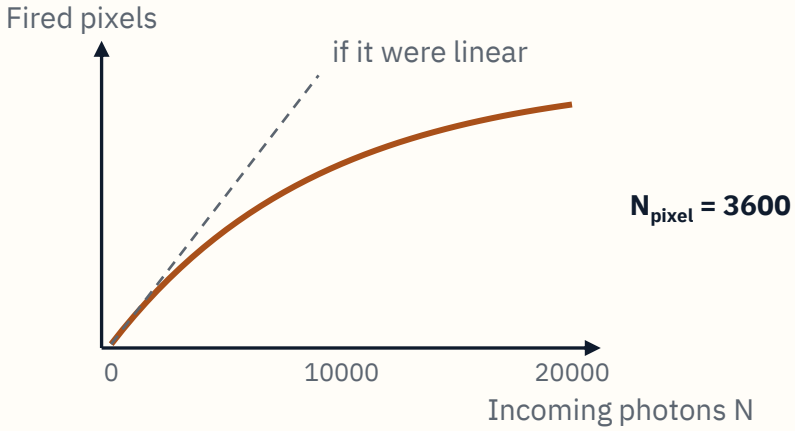
ΔV: overvoltage, 2 to 5 V; gain and PDE grow with ΔV, and so does the noise

Noise sources

- **Dark count:** Thermal firing without light, 50 to 100 thousand / s / mm² (room temperature). All of single-photon size, separated by a threshold.
- **Crosstalk:** A photon emitted by the avalanche fires a neighbouring pixel, 5% to 20%.
- **Afterpulse:** Trapped charge starts a delayed second avalanche, a few percent.

Saturation: The number of pixels is finite

$$N_{fired} = N_{pixel} \cdot (1 - e^{-N \cdot PDE/Npixel})$$



N_{fired}: Number of fired pixels (measured signal)

N, PDE: incoming photons and photon detection efficiency (≈ 0.4)

Example, 3600 pixels: 1000 photons: expected 400, fired 379 (5% loss). 10 000 photons: expected 4000, fired 2400 (40% loss), linearity is lost.

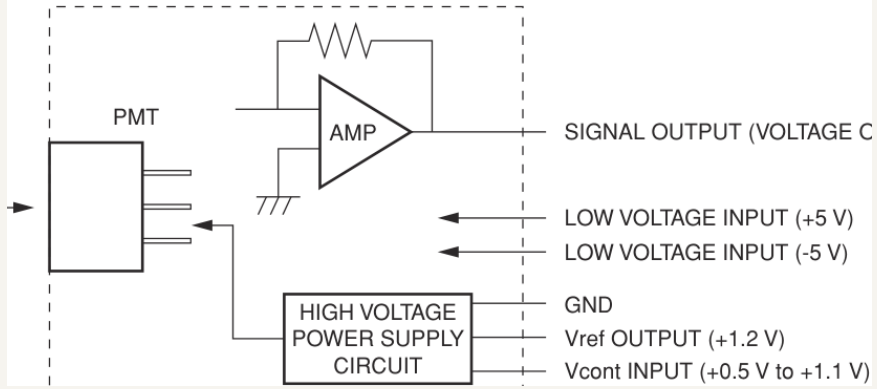
PMT and SiPM side by side

Property	PMT	SiPM	Verdict
Gain	10^5 to 10^7	10^5 to 10^6	Similar
Efficiency (QE / PDE)	25% to 30%	30% to 50%	SiPM sees more photons
Operating voltage	1000 to 2000 V	25 to 60 V	SiPM: simple electronics
Magnetic field	Does not work, needs shielding	Unaffected	SiPM inside a magnet
Size, robustness	glass tube of several cm, fragile	mm-sized chip, rugged	SiPM in tight spaces
Dark noise	100 to 1000 / s	10^5 / s / mm² (room temperature)	PMT is far quieter
Temperature	Weakly sensitive	Breakdown voltage drifts, needs correction	PMT is stable
Large area	A single tube up to 50 cm	With arrays, many channels	PMT for large volumes
Price	Hundreds of dollars	Tens of dollars	SiPM in many-channel systems

> **Trend:** PET, compact calorimeters and in-magnet detectors have moved to SiPMs. Large water Cherenkov and neutrino detectors still use PMTs.

PMT module: Hamamatsu H10722-110 datasheet

3: Schematic diagram



- **Module:** PMT + high voltage + amplifier in one box. The code uses the cathode and the gain of the PMT.
- **Simplicity:** No quantum efficiency in the code; every photon reaching the window counts as one photoelectron. N = photons reaching the PMT.
- **Signal:** N photons, gain G, R and C at the anode: V(t) by equation 8.13 (Leo), in step G.

Figure 1: Typical spectral response

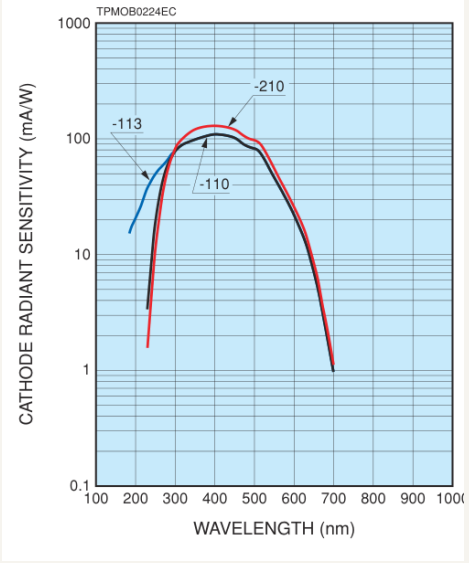
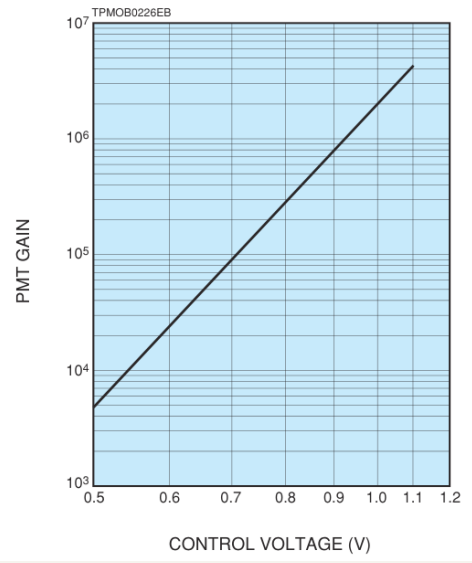


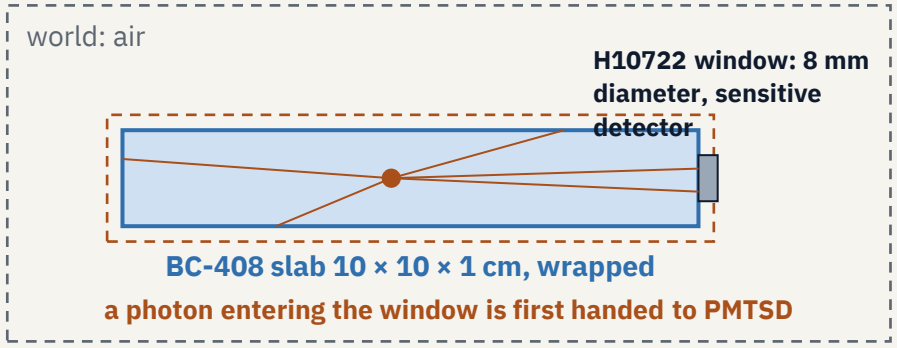
Figure 2: Typical gain



Magnitude: A 1 GeV muon sends ≈ 940 photons to the window. $G \cdot N \cdot e = 2 \times 10^6 \times 940 \times 1.6 \times 10^{-19}$ C = 300 pC. Across 50 Ω with τ = 5 ns the peak is of order 1.5 V.

- **Photocathode:** Super bialkali, 230 to 700 nm, peak at 400 nm
- **S(400 nm):** 110 mA/W, QE 34%
- **Effective area:** 8 mm diameter, borosilicate window
- **Gain:** 5×10^3 (0.5 V) to 2×10^6 (1.0 V). 1.0 V in the code
- **Anode circuit:** R = 50 Ω, C = 100 pF, τ = RC = 5 ns (in the code)

Step E: PMT window and sensitive detector



- **Window:** G4Tubs, radius 4 mm, half-thickness 0.75 mm. The cylinder axis is z, so it is rotated by 90° about y.
- **Glass:** Without RINDEX 1.47 the photon cannot pass from the slab into the glass; it dies at the boundary.
- **Sensitive detector:** PMTSD is created in ConstructSDandField and attached to the "PMT" logical volume.
- **Check:** Compile, run. The window exists, no hits yet: photons reaching the PMT 0.

```
// DetectorConstruction.cc
// --- STEP E: PMT window (leading // removed) ---
G4MaterialPropertiesTable* mptGlass = new
G4MaterialPropertiesTable();
mptGlass->AddProperty("RINDEX", energy, {1.47, 1.47, 1.47, 1.47});
glass->SetMaterialPropertiesTable(mptGlass);

G4Tubs* solidPMT = new G4Tubs("PMT", 0., 4 * mm, 0.75 * mm, 0., 360
* deg);
G4LogicalVolume* logicPMT = new G4LogicalVolume(solidPMT, glass,
"PMT");
G4RotationMatrix* rot = new G4RotationMatrix();
rot->rotateY(90 * deg);
new G4PVPlacement(rot, G4ThreeVector(5 * cm + 0.75 * mm, 0., 0.),
                  logicPMT, "PMT", logicWorld, false, 0);
// --- end of STEP E ---

void DetectorConstruction::ConstructSDandField()
{
  // --- STEP E: make the window a sensitive detector ---
  PMTSD* pmt = new PMTSD("PMT");
  G4SDManager::GetSDMpointer()->AddNewDetector(pmt);
  SetSensitiveDetector("PMT", pmt);   // name of the logical volume
  // --- end of STEP E ---
}
```

Step F: One hit per photon

hit and sensitive detector classes

```
// PMTHit.hh: the record of one photon (given)
class PMTHit : public G4VHit {
  G4double fEnergy;   // photon energy
  G4double fTime;     // arrival time
};
typedef G4THitsCollection<PMTHit> PMTHitsCollection;

// PMTSD.hh: sensitive detector (given)
class PMTSD : public G4VSensitiveDetector {
  void Initialize(G4HCofThisEvent*) override;   // per
event
  G4bool ProcessHits(G4Step*, G4TouchableHistory*)
override;
};
```

- **Hit:** The record of one photon. The hits of an event are kept in a collection named "PMT/PMTHits".
- **Killing:** The photon ends at the photocathode; it cannot bounce inside the glass and be counted twice.
- **Check:** Compile, run. Hits exist but are not read yet.

```
// PMTSD.cc: called for every step in the PMT volume
G4bool PMTSD::ProcessHits(G4Step* step, G4TouchableHistory*)
{
  // --- STEP F: create one hit per optical photon ---
  G4Track* track = step->GetTrack();
  if (track->GetDefinition() !=
      G4OpticalPhoton::OpticalPhotonDefinition()) return
false;

  PMTHit* hit = new PMTHit();
  hit->SetEnergy(track->GetKineticEnergy());    // energy
  hit->SetTime(step->GetPreStepPoint()->GetGlobalTime());
  fHits->insert(hit);

  track->SetTrackStatus(fStopAndKill);   // photocathode
  return true;
  // --- end of STEP F ---
  return false;
}
```

Step G: From hits to the output signal, V(t)

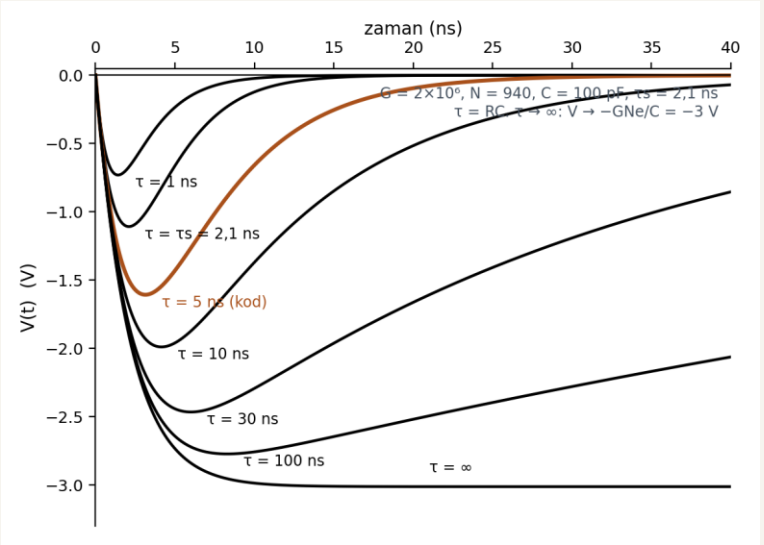
```
// EventAction.cc, end of event.  --- STEP G: from PMT hits to the output signal ---
G4HCofThisEvent* hce = event->GetHCofThisEvent();
if (fHCID < 0) fHCID = G4SDManager::GetSDMpointer()->GetCollectionID("PMT/PMTHits");
PMTHitsCollection* hits = (PMTHitsCollection*)hce->GetHC(fHCID);

G4int nphoton = hits->entries();            // N: photons reaching the PMT
G4double t0 = (*hits)[0]->GetTime();         // arrival of the first photon
for (G4int i = 1; i < nphoton; i++)
  if ((*hits)[i]->GetTime() < t0) t0 = (*hits)[i]->GetTime();
am->FillH1(3, nphoton);
for (G4int b = 0; b < 200; b++) {            // 0 - 50 ns
  G4double t = (b + 0.5) * (50. * ns / 200);
  am->FillH1(4, t + t0, PMTSignal(t, nphoton));   // V(t), summed over events
}
// --- end of STEP G ---

// given: output signal V(t), Leo equation 8.13 (mV, negative)
G4double PMTSignal(G4double t, G4int nphoton) {
  const G4double gain = 2.0e6, e_SI = 1.602e-19, R = 50., C = 100.0e-12;
  const G4double tau = R * C * s, tau_s = 2.1 * ns;      // RC = 5 ns, BC-408
  const G4double A = gain * nphoton * e_SI * R * 1000.;  // G N e R
  if (std::abs(tau - tau_s) > 1.0e-6 * ns)
    return -A / ((tau_s - tau) / ns) * (std::exp(-t / tau_s) - std::exp(-t / tau)) *
1.0e9;
  return -A / ((tau_s / ns) * (tau_s / ns)) * (t / ns) * std::exp(-t / tau_s) * 1.0e9;
}
```

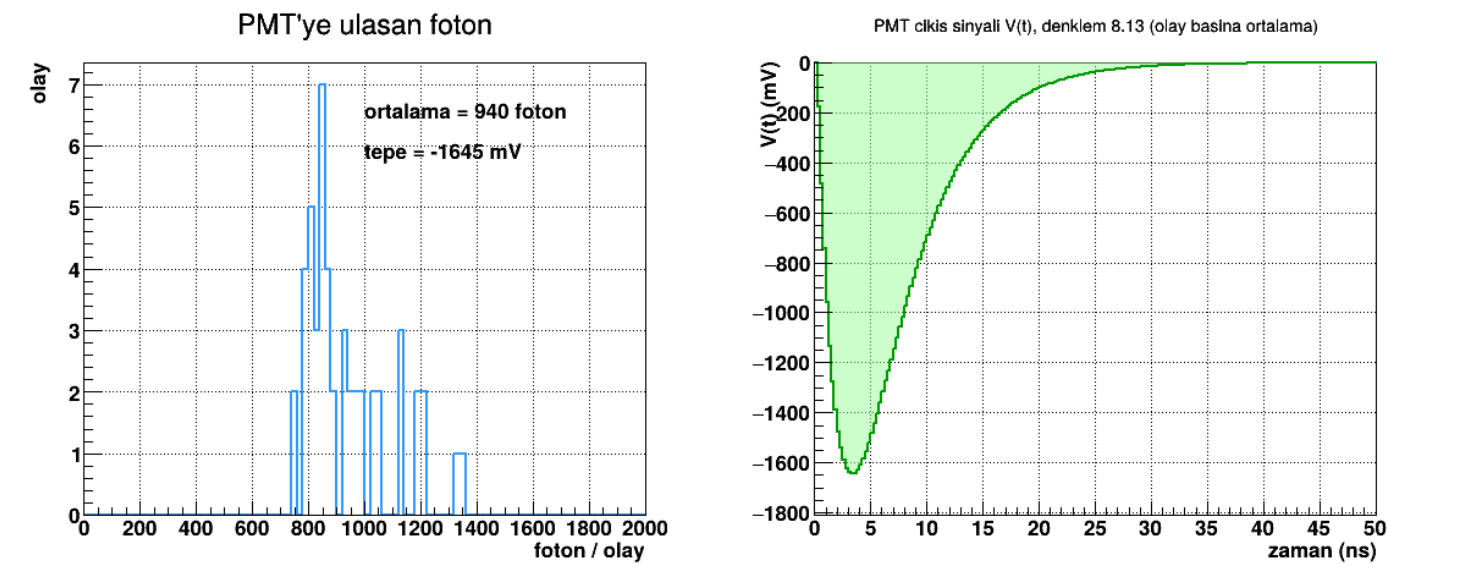
$$V(t) = \begin{cases} -\frac{GNeR}{\tau - \tau_s}\left[\exp\left(-\frac{t}{\tau_s}\right) - \exp\left(-\frac{t}{\tau}\right)\right] & \tau \neq \tau_s \\ \left(\frac{GNeR}{\tau_s^2}\right) t \exp\left(-\frac{t}{\tau_s}\right) & \tau = \tau_s \end{cases} \quad (8.13)$$

G gain, N photons reaching the PMT, e electron charge, R anode load resistor, τ = RC, τs scintillator decay time.
Figure 8.13: V(t) as τ changes.



Leo, equation 8.13 and Figure 8.13, with our parameters. In the code τ = 5 ns (orange).

Result: All steps from A to G open



Per event	Value
Photons produced	19 000
N reaching the window	940 (%5)
Gain (1.0 V)	2×10^6
Charge G · N · e	300 pC
τs, τ = RC	2.1 ns, 5 ns
Peak V(t)	−1.6 V, 3 ns

Reading the plot: Left: photons reaching the PMT per event. Right: output signal V(t), equation 8.13. A larger τ makes the pulse longer and lower; its area (the charge G · N · e) does not change.

- **Small window:** 8 mm diameter, 5% of the edge. Most photons are lost in the wrapping.
- **Shape:** Rise with τs, decay with τ = RC. The τ = 5 ns curve of Figure 8.13.
- **Try:** R, C, gain (PMTSignal), window diameter (step E). Quantum efficiency: multiply N by 0.3.

CLOSING

Where to study: Book chapters

Topic	Leo, Techniques for Nuclear and Particle Physics Experiments	Knoll, Radiation Detection and Measurement
Scintillation, organic and inorganic (02 to 04)	Chapter 7, Scintillation Detectors: 7.1 general characteristics, 7.2 organic, 7.3 inorganic crystals	Chapter 8, Scintillation Detector Principles: organic, inorganic, light yield and resolution
Light collection (05)	Chapter 7, Scintillation Detectors: light collection, light guide, wavelength shifter	Chapter 8, Scintillation Detector Principles: light collection
PMT (07)	Chapter 8, Photomultipliers: photocathode, dynodes, gain, time response, noise	Chapter 9, Photomultiplier Tubes and Photodiodes
PMT output signal (step G)	Chapter 8, Photomultipliers: equation 8.13 and Figure 8.13	Chapter 9: PMT pulse shape, anode circuit
SiPM (08)	None (the book predates the SiPM)	Chapter 9, Photodiodes and silicon photomultipliers (4th edition)
Geant4 optical photons (06, 10)	Geant4 Book For Application Developers: Optical Photon Processes	Datasheets: Saint-Gobain BC-408, Hamamatsu H10722

When studying for the exam, first the slides, then the relevant chapter. For anything unclear in the code, the README and the comment lines of the closed steps.

CLOSING

Summary

Scintillation	3% to 11% of the energy becomes light. $N = L_Y \cdot E$, decays exponentially with τ (NaI 230 ns, plastic 2 ns)
Organic	Molecular fluorescence, 1 to 10 ns. Plastic and liquid: fast, cheap, light
Inorganic	Band structure + activator (Tl, Ce), dense, high Z, lots of light. Gammas and calorimetry
Light collection	Total internal reflection ($\sin \theta_c = 1/n$), reflective wrapping, light guide, optical grease; 30% to 60% arrives
PMT	Photocathode (QE 25%) + 10 dynodes: $G = \delta^n \approx 10^6$, 1 to 2 kV, but sensitive to magnetic fields
SiPM	Thousands of pixels in Geiger mode. $G \approx 10^6$, 30 V, PDE 40%, but high dark count and saturation
Geant4	BC-408 slab: optics, wrapping. H10722 PMT: sensitive detector and hit, output signal V(t) simulation
Choice	Energy: NaI(Tl), LYSO. Timing: plastic. Inside a magnetic field and compact: SiPM. Large area: PMT

The chain: Deposited energy, photons, photoelectrons, multiplication, pulse. Its area gives the energy, its time the passage.